

# Self-Learning: 80386 Memory Management & MESI Cache Protocol MPCA

## Module 5

> **Definition:** The 80386 Memory Management Unit (MMU) provides two-stage address translation (**Segmentation & Paging**), while multiprocessor systems maintain cache consistency using the **MESI (Illinois) Protocol**.

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### 1. 80386 Protected Mode Memory Translation

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#### 1. Descriptor Tables (GDT, LDT, IDT):

- **GDT (Global Descriptor Table):** Holds segment descriptors accessible by all system tasks.

- **L**

#### 2. Two-Level Paging Scheme (4 KB Pages):

A 32-bit Linear Address is split into:

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### 2. MESI Cache Coherence Protocol

In symmetric multiprocessing (SMP) systems with private L1/L2 caches, every cache line exists in one of **four MESI states**:

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M - MC

State Transition Summary:

- \*\*Read Hit:\*\* Stays in M, E, or S state.

- \*\*W

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3. Quick Recall Flow

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